

Research Paper

Emergence of high-risk multidrug-resistant *Enterococcus faecalis* CC2 (ST181) and CC87 (ST28) causing healthcare-associated infections in IndiaChandrabhan Rao^a, Benu Dhawan^{a,*}, Sreenivas Vishnubhatla^b, Arti Kapil^a, Bimal Das^a, Seema Sood^a^a Department of Microbiology, All India Institute of Medical Sciences, New Delhi 110029, India^b Department of Biostatistics, All India Institute of Medical Sciences, New Delhi 110029, India

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ABSTRACT

High-risk hospital-associated multidrug-resistant (MDR) *Enterococcus faecalis* clonal complexes (CCs) such as CC2 and CC87 are enriched with virulence determinants that help to accumulate, colonize, and cause serious nosocomial infections. The aim of this study was to establish the epidemiology and clonal composition of 134 clinical *E. faecalis* isolates and to link molecular typing data with antimicrobial resistance and virulence determinants. All isolates were identified by conventional methods and confirmed by polymerase chain reaction (PCR) (*16sRNA* gene and *ddl* genes of *E. faecalis*/*E. faecium*) in 5-years. Disc diffusion test was performed on all strains. We screened all *E. faecalis* for *aac(6')-aph(2'')*, *vanA*, and *vanB* resistance genes, and aggregation substance-*asa1*, cytotoxin-*cylA*, collagen-binding protein-*ace*, enterococcal surface protein-*esp*, gelatinase-*gelE*, and hyaluronidase-*hyl* virulence genes by PCR. Representative isolates of *E. faecalis* were characterized by pulsed-field gel electrophoresis (PFGE) and multi-locus sequence typing (MLST). Out of 539 patients with enterococcal infections, 134 (24.9%) had *E. faecalis* infections, 366 (67.9%) had *E. faecium* infections, and 39 (7.2%) had infections due to other enterococcal species. Of the 134 isolates, 79.1% and 61.9% isolates were high-level gentamicin resistant (HLGR) and MDR. In multivariate analysis, independent predictor for infection due to MDR *E. faecalis* strains was a surgical intervention (OR 2.41, 95% CI 1.17–4.96, $P = 0.017$). Overall, the observed rate of in-hospital mortality was 11.9%. The *gelE*, *asa1*, *ace*, *cylA*, *esp* and *hyl* genes were detected in 87.3%, 78.4%, 54.5%, 53.7%, 36.6% and 3.0%, respectively in *E. faecalis* isolates. The *asa1*, *cylA*, and *gelE* genes were significantly correlated with MDR *E. faecalis*. The PFGE analysis showed 28 clones with four major clones. MLST analysis revealed two sequence types-ST28 (CC87) and ST181 (CC2). This is the first Indian report on the emergence of the high-risk hospital-associated worldwide-disseminated ST28 (CC87) and ST181 (CC2), which have enriched with multiple virulence determinants and resistance to antibiotics, particularly ampicillin. This report indicates serious health concern and calls for on-going surveillance, close monitoring, and improved infection control procedures to stop further spread of these isolates.

1. Introduction

Enterococcus faecalis is a major cause of enterococcal nosocomial infections. *E. faecalis* isolates acquire and transfer virulence and antibiotic resistance determinants, which often create a variety of therapeutic dilemmas. The emergence of multidrug-resistant (MDR) *E. faecalis* has become a major public health threat, as it has limited the effective antimicrobial agents available to treat infections caused by this bacterium. *E. faecalis* infections are often present as a notable therapeutic challenge because of the ease of acquisition and aforementioned transfer of antimicrobial drug resistance, particularly their

high-level resistance to aminoglycoside and glycopeptide resistance, along with their intrinsic resistance to various antibiotics (Arias and Murray, 2012; Kuch et al., 2012; McBride et al., 2007; Weiner et al., 2016).

Synergistic killing effects against clinical *E. faecalis* isolates have been predicted to treat such serious infections by employing combination therapy with the administration of a high-level aminoglycoside, such as high-level gentamicin and cell wall inhibitors particularly penicillin, ampicillin or vancomycin. However, such synergistic effects are eliminated due to the acquisition of the most clinically relevant *aac(6)-aph(2'')* gene of bi-functional aminoglycoside modifying enzyme (AME)

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in *E. faecalis*, which is responsible for resistance to all clinically relevant aminoglycoside except streptomycin (Arias and Murray, 2008; Chow, 2000). Rare resistance to penicillin or ampicillin in *E. faecalis* strains has also been on the rise due to the increases in the quantity of low-affinity penicillin-binding protein 4 (Pbp4) or with amino acid substitutions in Pbp4 (Moon et al., 2018). Glycopeptide resistance increases because of the acquisition of van genes, mainly *vanA* and *vanB* genes by *E. faecalis* isolates (Arias and Murray, 2012).

The ability of *E. faecalis* isolates to cause serious infections in hospitalized patients is directly linked to the *E. faecalis* pathogenicity islands (PAIs) which encode multiple virulence factors, including the cytotoxin toxin (*cylA*), a gelatinase (*gelE*), enterococcal surface protein (*esp*), aggregation substance (*asaI*), collagen-binding proteins (*ace*), and hyaluronidase (*hyl*). These factors make it possible for the organism to remain in the hospital environment or to change its relationship with the host to survive many of the host defenses (Billström et al., 2008; Djahmi et al., 2012; Hasani et al., 2012; Rice et al., 2018; Shankar et al., 2002).

Pulsed-field gel electrophoresis (PFGE) and Multi-locus sequence typing (MLST) have been employed as a common molecular tool in many epidemiological studies. These studies show *E. faecalis* as major nosocomial pathogens with specific hospital-associated lineages or clonal complexes that accumulate antimicrobial resistance and virulence determinants (Kuch et al., 2012; McBride et al., 2007). In previous studies, *E. faecalis* MLST revealed the existence of some distinct genetic sub-populations called clonal complex (CC)2, CC16, CC21, CC30, CC40, and CC87 being found in most hospital-related infections and outbreaks worldwide (Kuch et al., 2012; McBride et al., 2007). The *E. faecalis* strains belong to CC2, CC9, and CC87, which are considered to be high-risk CCs causing infections in hospitalized patients (Kuch et al., 2012; Ruiz-Garbajosa et al., 2006). *E. faecalis* MLST plays a key role in the study of the clinical epidemiology of *E. faecalis* clones that routinely cause hospital-associated infections (HAIs) (Kawalec et al., 2007; Kuch et al., 2012; McBride et al., 2007; Ruiz-Garbajosa et al., 2006).

However, to the best of our knowledge, there is no Indian perspective on the molecular epidemiology and clonal composition of *E. faecalis* infecting humans. To fill this gap, we collected *E. faecalis* isolates from nosocomial infections in an Indian tertiary hospital and analyzed their antimicrobial susceptibility, putative virulence determinants, and clonal relatedness by PFGE and MLST.

2. Materials and methods

2.1. Study setting and design

This study was carried out at 2500 inpatient bed tertiary care referral and teaching hospital in New Delhi. Ethical approval for this work was obtained from the Ethics Committee of the All India Institute of Medical Science (AIIMS), New Delhi (No.: IEST/T-151/2010) for this work. We collected data on the demographic characteristics of patients, underlying diseases, antimicrobial use, receipt of immunosuppressive drugs, presence of invasive devices at the time of *E. faecalis* isolation, location in the hospital, including intensive care unit (ICU) stay, Length of hospital stay (LOS), Charlson Co-morbidity Index (CCI) score to measure the strength of co-morbidities (Charlson et al., 1987) and McCabe Scores for the severity of the underlying disease (McCabe and Jackson, 1962) to evaluate the risk factors for MDR *E. faecalis* vs. Non-MDR *E. faecalis*. All strains of *E. faecalis* were isolated from the blood, skin, and soft tissue (SSTI) and urine samples routinely received in the Microbiology laboratory from admitted patients during 2010–2014.

2.2. Bacterial identification

All *E. faecalis* and *E. faecium* isolates initially identified with standard biochemical reactions and further confirmed by polymerase chain

reaction (PCR) for the genus level by the 16S rRNA gene (van de Klundert and Vliegthart, 1993) and species level by D-Ala-D-Ala ligase genes for *E. faecalis* and *E. faecium* (Depardieu et al., 2004).

2.3. Antimicrobial susceptibility testing

The antimicrobial resistance profile of all *E. faecalis* strains was determined by the disc diffusion method (Kirby–Bauer) and interpreted as CLSI guideline (Clinical and Laboratory Standards Institute, 2012), against a panel of 12 antimicrobial agents: ampicillin (10 µg), chloramphenicol (30 µg), ciprofloxacin (5 µg), erythromycin (15 µg), gentamicin (120 µg) to detect high-level aminoglycoside resistance (HLGR), fosfomycin (200 µg) supplemented with 50 µg of glucose-6-phosphate, linezolid (30 µg), rifampin (5 µg), tetracycline (30 µg), tigecycline (15 µg), teicoplanin (30 µg) and vancomycin (30 µg). We used the broth dilution method to test the minimum inhibitory concentration (MIC) of teicoplanin and vancomycin, and considered vancomycin MIC: susceptible, ≤ 4 µg/ml; intermediate, 8–16 µg/ml; resistant, ≥ 32 µg/ml. However, teicoplanin MIC was considered as: susceptible, ≤ 8 µg/ml; intermediate, 16 µg/ml; resistant, ≥ 32 µg/ml. We used *E. faecalis* ATCC 29212 and *Staphylococcus aureus* ATCC 25923 strains as reference quality control in the broth dilution method and disc diffusion method (Clinical and Laboratory Standards Institute, 2012). *E. faecalis* MDR was defined as acquired non-susceptibility to at least one agent in three or more categories of antimicrobial (Magiorakos et al., 2012).

2.4. Detection of antibiotic resistance and virulence determinants

We used multiplex PCR to detect *aac(6′)-aph(2′)* gene for HLGR and *vanA* and *vanB* genes for vancomycin resistance genetic determination as described previously (Ramos-Trujillo et al., 2003; Yean et al., 2007). We used another multiplex PCR to detect *ace*, *asaI*, *gelE*, *cylA*, *esp*, and *hyl* genes as virulence determinants of *E. faecalis* as described previously (Billström et al., 2008).

2.5. Pulsed-field gel electrophoresis (PFGE)

We performed PFGE typing on 103 strains of *E. faecalis* isolated from different samples viz. blood (*n* = 30), SSTI (*n* = 35) & urine (*n* = 38) following total DNA digestion by *Sma*I. Representative strains of each antimicrobial profile of *E. faecalis* were included in the PFGE analysis. We performed PFGE on HLGR (*n* = 89) and non-HLGR (*n* = 14) *E. faecalis* isolates as previously described (Dhawan et al., 2015; Udo et al., 2009).

Of the 103 isolates from *E. faecalis*, 30 were isolated from blood, 35 from SSTI, and 38 from urine samples. PFGE typing was done by CHEF-Mapper apparatus (Bio-Rad Laboratories, Hercules, CA). The National Collection of Type Culture (NCTC) 8325 strain was loaded to all PFGE gels as a reference size marker. We set the machine program for electrophoresis as follows: ramped pulsed times 5.3–34.9 s for 20 h, temperature 14 °C, and the voltage 6.0 (V/Cm). The InfoQuest™ FP Software v. 5.4 (Bio-Rad Laboratories, USA) has been applied to compare and group PFGE patterns. Composite trees was generated by band-based similarity coefficient (Dice) and unweighted pair group method with arithmetic mean (UPGMA) with 1.0% optimization and 1.0% band position tolerance. To define PFGE patterns, ≥ 75% similarity coefficient was considered (Dhawan et al., 2015; Tzavaras et al., 2012).

2.6. Multi-locus sequence typing (MLST)

Representative MDR *E. faecalis* isolates were selected from different PFGE types (major and sporadic clones) for MLST analysis. We performed MLST as described previously (Kuch et al., 2012); allele numbers and sequence types (STs) were assigned using the *E. faecalis* MLST database (www.pubmlst.org).

2.7. Statistical analysis

All statistical analyses were performed with Stata v. 12.1 (StataCorp., USA). Odds ratios (ORs) were calculated from the logistic regression analysis for MDR *E. faecalis* vs. Non-MDR *E. faecalis* along with 95% Confidence Intervals (CIs). We applied univariate logistic regression analysis to identify the independent predictors associated with MDR *E. faecalis* vs. Non-MDR *E. faecalis*. A Chi-squared test was used to compare the resistance patterns of *E. faecalis* and the correlation between the occurrence of virulence genes and antibiotic resistance. All *P* values were two-tailed and a *P*-value < 0.05 was considered statistically significant.

3. Results

3.1. Distribution of *E. faecalis* isolates in clinical samples

A total of 539 hospitalized patients with enterococcal infections were identified during the 5-year study period. Out of 539 patients, 134 (24.9%) had *E. faecalis* infections, 366 (67.9%) had *E. faecium* infections, and 39 (7.2%) had infections due to other enterococcal species. Thus, 134 *E. faecalis* strains were isolated from hospitalized patients during the 5-year study period. Of the 134 *E. faecalis* isolates, 51 (38.1%) were isolated from skin and soft tissue, 46 (34.3%) from urine, and 37 (27.6%) from blood samples.

3.2. Antimicrobial susceptibility results, high-level gentamicin resistance (HLGR), and multidrug-resistant (MDR)

Most *E. faecalis* strains were resistant to erythromycin (88.8%, *n* = 119), ciprofloxacin (82.1%, *n* = 110), gentamicin (79.1%, *n* = 106), tetracycline (69.4%, *n* = 93), rifampicin (50.0%, *n* = 67), and chloramphenicol (41.0%, *n* = 55). However, only 27.6% (*n* = 37) of *E. faecalis* were resistant to ampicillin. All strains were uniformly susceptible to linezolid, tigecycline, fosfomycin, vancomycin, and teicoplanin. All *E. faecalis* isolates were also susceptible to vancomycin and teicoplanin by the broth dilution method as well as by *vanA* and *vanB* genes. HLGR was detected by *aac(6')-aph(2'')* gene in 79.1% (*n* = 106) of *E. faecalis* isolates. However, all strains of *E. faecalis* were negative for *vanA* and *vanB* genes. MDR was detected in 61.9% (*n* = 83) of the *E. faecalis* isolates.

3.3. Risk factors associated with multidrug-resistant (MDR) *E. faecalis*

Of the 134 isolates, 61.9% (*n* = 83) MDR *E. faecalis* and 38.1% (*n* = 51) non-MDR *E. faecalis* strains were identified. The demographic characteristics of patients with infection due to MDR *E. faecalis* and non-MDR *E. faecalis* strains are shown in Table 1. In a multivariate analysis, the independent predictor for infection due to MDR *E. faecalis* strains was surgical intervention (OR 2.41, 95% CI 1.17–4.96, *P* = 0.017). Overall, the mortality rate was observed in 11.9% in admitted patients infected with *E. faecalis* isolates. There was no significant difference between the other outcomes of patients infected with MDR *E. faecalis* strains compared to non-MDR *E. faecalis* strains.

3.4. Prevalence of virulence determinants

In this study, the percentage of enterococci harboring at least one virulence determinant was 94.0% (*n* = 126) and 82.1% (*n* = 110) of *E. faecalis* isolates carrying two or more virulence genes. The *gelE*, *asa1*, *ace*, *cylA*, *esp* and *hyl* genes detected at 87.3%, 78.4%, 54.5%, 53.7%, 36.6% and 3.0%, respectively. The virulence determinant of 51 Non-MDR *E. faecalis* and 83 MDR *E. faecalis* strains is summarized in Table 2. Virulence genes-*asa1*, *cylA*, and *gelE* were significantly associated with MDR *E. faecalis* compared to Non-MDR *E. faecalis* strains.

3.5. Correlation between the presence of virulence genes and antibiotic resistance

The significant correlation between the presence of virulence genes and antibiotic resistance is shown in Table 3. There was no significant correlation between the presence of the *esp* gene and antibiotic resistance observed in all antibiotics tested data not shown here. No statistical analyses was performed on the correlation between the more rare virulence gene (*hyl*) and antibiotic resistance of the strains of *E. faecalis* as the numbers were too less.

There was no significant difference in the distribution of virulence genes in *E. faecalis* strains isolated from bloodstream infection (BSI), skin and soft tissue infection (SSTI), and urinary tract infections (UTI).

3.6. Molecular typing by pulsed-field gel electrophoresis (PFGE)

The *SmaI* macrorestriction fragment profiles of 103 *E. faecalis* isolates revealed 28 clones, named from 1 to 28, with four major clusters predominating (clone 12, *n* = 38, 36.9%; clone 3, *n* = 9, 8.7%; clone 16, *n* = 9, 8.7%; clone 11, *n* = 6, 5.8%) and together representing 60.2% (*n* = 62) of selected *E. faecalis* strains for PFGE. Each major clone consisted of 6–33 subtypes. Major clones 12, 3, 16, and 11 were further classified into 33, 9, 8, and 6 subtypes. Of the sporadic PFGE types (*n* = 41), 5 clones consisted of 3 strains each, 7 clones consisted of 2 strains each, and 12 clones consisted of only one strain (Fig. 1).

3.7. Molecular typing by multi-locus sequence typing (MLST)

The MLST results of 10 *E. faecalis* strains showed two different sequence types (STs)- ST28 and ST181. The distribution of STs was as follows: ST28 (*n* = 6); and ST181 (*n* = 4) belonging to CC87 and CC2 respectively (Table 4). The ST28 and ST181 nucleotide sequences of seven genes alleles identified in this study were deposited with GenBank and assigned accession numbers MT825643 to MT825670.

3.7.1. CC2-ST181

Major PFGE clone 12 comprised four isolates of ST181. All ST181 isolates were resistant to ciprofloxacin, erythromycin, gentamicin, and tetracycline. However, 75% of the ST181 isolates were chloramphenicol and rifampicin resistant. All strains of this clone were positive for *asa1* and *gelE* genes. Their virulence profiles were *asa1-gelE* (25%, *n* = 1), *esp-asa1-cylA-gelE-ace* (25%, *n* = 1), *asa1-gelE-ace* (25%, *n* = 1), *asa1-cylA-gelE* (25%, *n* = 1). All ST181 strains recovered from hospitalized patients of surgical wards only (Table 4).

3.7.2. CC87-ST28

All the ST28 isolates were resistant to ciprofloxacin, erythromycin and tetracycline. However, 83% of ST28 isolates were resistant to gentamicin and rifampicin, 50% to chloramphenicol and 33% to ampicillin. Their virulence profiles were *asa1-cylA-gelE-ace* (33.4%, *n* = 2), *esp-asa1-cylA-gelE-ace* (33.4%, *n* = 2), *esp-hyl-asa1-cylA-gelE-ace* (16.7%, *n* = 1), *gelE* alone (16.7%, *n* = 1). These ST28 strains were recovered from hospitalized patients of Medical wards (50%, *n* = 3) and ICUs (50%, *n* = 3) only (Table 4).

4. Discussion

This is the first comprehensive study from India on sample wise distribution, antimicrobial resistance, putative virulence determinants, and molecular epidemiology of *E. faecalis* strains isolated from clinical samples. Historically, the ratio of infection due to *E. faecalis* was 10:1 compared to other *Enterococcus* species. *E. faecalis* is still a dominant species of *Enterococcus* in different studies reported from Algeria (64%), Iran (69%), and the USA (48.9%) (Djahmi et al., 2012; Hasani et al., 2012; Sievert et al., 2013). In contrast to these reports, the increase in the proportion of *E. faecium* from the USA (38–75%), China (58.7%),

Table 1

Comparison of demographics, clinical characteristics, and outcomes of multidrug-resistant (MDR) and non-MDR *E. faecalis* strains isolated from blood, skin and soft tissue and urine samples ($n = 134$).^a

Characteristics	Non-MDR <i>E. faecalis</i> ($n = 51$)	MDR <i>E. faecalis</i> ($n = 83$)	OR	95% CI	P
Age, mean $\pm$ S.D	45.31 $\pm$ 21.74	38.15 $\pm$ 22.39	0.99	0.97–1.00	0.07
Sex					
Male	22 (43.1)	39 (47.0)	1		0.66
Female	29 (56.9)	44 (53.0)	0.86	0.42–1.73	
Hospitalization ward					
Medical	12 (23.5)	18 (21.7)	1		
Surgical	27 (52.9)	48 (57.8)	1.18	0.50–2.83	0.70
ICU	12 (23.5)	17 (20.5)	0.94	0.33–2.67	0.91
LOS before culture (days), mean $\pm$ S.D	11.00 $\pm$ 13.95	14.93 $\pm$ 13.30	1.02	0.99–1.05	0.11
0–10	33 (66.0)	36 (44.4)	1		
11–20	10 (20.0)	24 (29.6)	2.20	0.92–5.28	0.08
> 20	7 (14.0)	21 (25.9)	2.75	1.03–7.31	0.04
Unknown	1 (2.0)	2 (2.4)	–	–	..
McCabe Score, mean $\pm$ S.D	1.20 $\pm$ 0.85	1.14 $\pm$ 0.87	0.93	0.62–1.40	0.74
Charlson's comorbidity index, mean $\pm$ S.D	1.55 $\pm$ 1.01	1.41 $\pm$ 1.23	0.90	0.66–1.22	0.50
Risk factors associated					
Previous hospitalization (< 6 months)	9 (17.6)	21 (25.3)	1.58	0.66–3.79	0.30
ICU admission before culture	25 (49.0)	49 (59.0)	1.5	0.74–3.02	0.26
Inter-hospital transfer	2 (3.9)	9 (10.8)	2.98	0.62–14.4	0.17
Intra-hospital transfer	33 (64.7)	62 (74.7)	1.61	0.76–3.44	0.22
Intravenous catheter in use at time of culture	47 (92.2)	78 (94.0)	1.33	0.34–5.19	0.68
Urinary catheter in use at time of culture	29 (56.9)	57 (68.7)	1.66	0.81–3.43	0.17
Prosthetic devices	2 (3.9)	5 (6.0)	1.57	0.29–8.41	0.6
Endotracheal intubations	14 (27.5)	21 (25.3)	0.9	0.41–1.97	0.78
Underlying diseases/conditions					
Haematological malignancies	2 (3.9)	2 (2.4)	0.61	0.08–4.43	0.62
Solid Tumour	7 (13.7)	20 (24.1)	2	0.78–5.12	0.15
Gastrointestinal disease	93 (63.7)	22 (53.2)	0.65	0.33–1.26	0.20
Hepatitis	1 (2.0)	1 (1.2)	0.61	0.04–9.97	0.73
Diabetes mellitus	12 (23.5)	14 (16.9)	0.66	0.28–1.57	0.35
Renal disease	10 (19.6)	9 (10.8)	0.5	0.19–1.33	0.16
Rheumatic/connective tissue disease	5 (9.8)	10 (12.0)	1.26	0.41–3.92	0.69
Endocarditis	1 (2.0)	2 (2.4)	1.24	0.11–14.0	0.87
Neurological Disease	8 (15.7)	15 (18.1)	1.19	0.46–3.03	0.72
Genito-urinary tract disease	6 (4.1)	4 (8.5)	2.17	0.59–8.05	0.24
Immune deficiency	9 (17.6)	23 (27.7)	1.79	0.75–4.25	0.19
Sepsis	4 (7.8)	5 (6.0)	0.75	0.19–2.95	0.68
Corticosteroids	8 (15.7)	12 (14.5)	0.91	0.34–2.40	0.85
Surgical interventions	23 (45.1)	55 (66.3)	2.39	1.17–4.89	0.02
Recent Antimicrobial Therapy					
Penicillins ^b	16 (31.4)	20 (24.1)	0.69	0.32–1.51	0.36
Aminoglycosides	25 (49.)	43 (51.8)	1.12	0.56–2.25	0.75
Metronidazole	16 (31.4)	27 (32.5)	1.06	0.50–2.23	0.89
Fluoroquinolones	20 (39.2)	27 (32.5)	0.75	0.36–1.54	0.43
Linezolid	4 (7.8)	8 (9.6)	1.25	0.36–4.39	0.72
Glycopeptides	7 (13.7)	9 (10.8)	0.76	0.27–2.20	0.62
Clindamycin	2 (3.9)	0 (0.0)	–	–	–
Carbapenem	4 (7.8)	9 (10.8)	1.43	0.42–4.91	0.57
Cephalosporin (3rd generations)	38 (74.5)	57 (68.7)	0.75	0.34–1.64	0.47
Total number of antibiotics received, mean $\pm$ S.D	3.02 $\pm$ 1.79	3.01 $\pm$ 1.66	0.99	0.81–1.22	0.98
Monotherapy	5 (9.8)	11 (13.3)	1		
Dual combination therapy	15 (29.4)	24 (28.9)	0.73	0.21–2.51	0.61
Triple or multiple combination therapy	30 (58.8)	47 (56.6)	0.71	0.22–2.25	0.56
No antibiotic therapy	1 (2.0)	1 (1.2)	0.45	0.02–8.83	0.60
Outcome					
Mortality	8 (15.7)	8 (9.6)	0.57	0.20–1.64	0.3
LOS total (days), mean $\pm$ S.D	19.22 $\pm$ 16.01	24.21 $\pm$ 16.63	1.02	0.99–1.04	0.10

Data are no. (%) of patients, unless otherwise specified.

P values ≤ 0.05 are highlighted in bold.

^a Abbreviations: OR, Odds ratio; CI, confidence interval; ICU intensive care unit; LOS, length of hospital stay.

^b Penicillins include β -Lactams- β -Lactamase inhibitor combinations.

and our hospital (67.9%) has been reported (Hidron et al., 2008; Jia et al., 2014).

In this study, *E. faecalis* strains were predominantly isolated from the skin and soft tissue infections (38.1%), followed by urine samples (34.3%) and blood samples (27.6%). Similarly, Brazilian authors isolated *E. faecalis* predominantly from wound samples (41.5%), followed by urine samples (28.4%) and blood (6.7%) (Conceição et al., 2011). Another Saudi Arabia study showed that the distribution of *E. faecalis*

strains were 35.5% in wounds, 19.3% in urine, and 13.2% in the blood (Salem-Bekhit et al., 2012). In contrast to our study, *E. faecalis* strains were predominantly isolated from central line-associated bloodstream infection (CLABSI) (48.9%) followed by catheter-associated urinary tract infection (CAUTI) (27.7%), surgical site infection (22.6%), and ventilator-associated pneumonia (0.8%) in the National Healthcare Safety Network (NHSN) (Sievert et al., 2013). Extensive presence of the *E. faecalis* isolates in clinical samples of hospitalized patients in our

Table 2

Distribution and comparative analysis of virulence genes between MDR and non-MDR *E. faecalis* strains (n = 134).

Virulence genes	<i>E. faecalis</i> (n = 134)	Non-MDR <i>E. fs</i> (n = 51)	MDR <i>E. fs</i> (n = 83)	P
Cell surface determinants				
<i>esp</i>	49 (36.6)	15 (29.4)	34 (41.0)	0.18
<i>asa1</i>	105 (78.4)	31 (60.8)	74 (89.2)	< 0.001
<i>ace</i>	73 (54.5)	23 (45.1)	50 (60.2)	0.09
Secreted factors				
<i>cylA</i>	72 (53.7)	18 (35.3)	54 (65.1)	0.001
<i>gelE</i>	117 (87.3)	40 (78.4)	77 (92.8)	0.01
<i>hyl</i>	4 (3.0)	2 (3.9)	2 (2.4)	0.62

All P values determined using the χ^2 test. P values ≤ 0.05 are highlighted in bold.

study and other studies confirms its active role in causing nosocomial infections such as skin and soft tissue infections (SSTIs), urinary tract infections (UTIs), and blood stream infections (BSIs).

Comparing our previous and current studies, there is a significant increase in trend (10.0% to 38.1%) of the skin and soft tissue infections caused by *E. faecalis* isolates in our hospital (Mathur et al., 2003). Similar to our findings, vancomycin, teicoplanin, linezolid, and tigecycline were highly effective against *E. faecalis* strains with resistance < 3%, and these drugs remained globally available as treatment options for related infections (Brandon and Dowzicky, 2013; Djahmi et al., 2012; Kuch et al., 2012). However, recently, we reported that vancomycin-resistant *E. faecium* (VREfm) was detected in 42.2% of *E. faecium* strains isolated from blood-stream infections (Rao et al., 2020).

The recommended combination therapy for most *E. faecalis* infections is high-level aminoglycoside (such as gentamicin) and cell wall inhibitors, particularly ampicillin and vancomycin due to their synergistic therapeutic effects. This synergistic therapeutic activity between high-level aminoglycosides and cell wall inhibitors is eliminated due to the presence of several aminoglycoside modifying enzymes (AMEs). However, most clinically relevant AME in *E. faecalis* is the bifunctional aminoglycoside-modifying enzyme that encodes the *aac* (6)-*aph*(2") gene and is responsible for high-level gentamicin resistance, including resistance to all clinically relevant aminoglycosides except streptomycin (Arias and Murray, 2008; Chow, 2000). The prevalence of HLGR (43.3%) in European countries was almost similar to that of high-level streptomycin-resistant (HLSR) (40.0%) (Kuch et al., 2012). Therefore, instead of the detecting all AMEs and HLSR, we only preferred to detect *aac*(6)-*aph*(2") gene and HLGR for this study.

The production of beta-lactamase and overproduction of penicillin-binding proteins (PBPs) are the two main mechanisms for resistance to beta-lactam (such as ampicillin) in *E. faecalis*. However, mutations or

modifications in Pbp4, or with amino acid substitutions in Pbp4 at both positions 520 and 605 are associated with increased resistance to penicillin/ ampicillin which is a potent therapeutic option against MDR *E. faecalis* (Moon et al., 2018). Still, ampicillin is a choice of antibiotic for the treatment of penicillin-susceptible VRE infection (Brandon and Dowzicky, 2013).

Brandon and Dowzicky reported significantly increased susceptibility to ampicillin and penicillin in gram-positive organisms collected from paediatric patients worldwide during their 2004 to 2011 study period. According to their study, resistance of *E. faecalis* isolates were < 3% (Brandon and Dowzicky, 2013). In another study conducted by Salem-Bekhit and colleagues in Saudi Arabia, 15.7% of *E. faecalis* isolates were resistance to ampicillin with MIC > 16 μ g/ml (Salem-Bekhit et al., 2012). Comparatively, we documented a high proportion of *E. faecalis* resistant to many antibiotics such as ampicillin (27.6%), HLGR (79.1%), ciprofloxacin (82.1%), and chloramphenicol (41.0%). Our results indicate that *E. faecalis* strains have increased resistance to antibiotics due to hospital adaptation and cause high morbidity and mortality in infected patients. Also, we noticed the significant increase in resistance to high-level gentamicin (69.0% to 79.1%), and erythromycin (83.3% to 88.8%) in our hospital over the last decade (Mohanty et al., 2005), has reduced the therapeutic options available to our patients and the most troubling aspect is that this resistance determinant is transferable to the bacteria through plasmids (Watanabe et al., 2009). Two studies from Europe showed that MDR in *E. faecalis* isolates were 59.7% and 59.0%, which were close to our MDR report in 61.9% of hospital *E. faecalis* isolates (Kawalec et al., 2007; Kuch et al., 2012). The rates of resistance to all antibiotics were significantly greater among MDR *E. faecalis*, as expected. The significant presence of MDR strains of *E. faecalis* during the surgical intervention procedure complicates treatment options and pointed to a modified infection control procedure to control its spread.

In a population-based study from Canada, the observed mortality rate was 17.0% in patients infected with *E. faecalis*, compared to 11.9% in our study (Billington et al., 2014). However, Hayakawa and colleagues reported 21% in-hospital mortality in patients infected with vancomycin-resistant *E. faecalis* (Hayakawa et al., 2012). Differences in the mortality rate may be possible due to variation in case mixes and geographical location in other studies.

A study from Iran showed that the rates of *gelE*, *asa1*, *ace*, *cylA*, and *esp* genes were 90.8%, 71.7%, 73.7%, 32.2%, and 45.4%, respectively in *E. faecalis* (Hasani et al., 2012). However, the *gelE*, *asa1*, *cylA*, *esp* and *hyl* genes were detected in 73.8%, 37.5%, 1.0%, 22.5%, and 0.0%, respectively in another Algerian study (Djahmi et al., 2012). Similarly, we found a high rate of *gelE* (87.3%) and *asa1* (78.4%), a moderate rate of *ace* (54.5%), *cylA* (53.7%), and *esp* (36.6%), and the low prevalence of *hyl* (3.0%) virulence genes. However, Zalipour and his colleagues reported high prevalence of the *gelE* (92.4%), *asa1* (90.6%), *ace* (86.8%), *esp* (75.5%) and *hyl* (18.9%) genes, compared to our results (Zalipour

Table 3

Correlation between presence of virulence genes and antimicrobial resistance patterns of *E. faecalis* strains (n = 134).

Antibiotics	<i>asa1</i> negative (n = 29)	<i>asa1</i> positive (n = 105)	P ¹	<i>cylA</i> negative (n = 62)	<i>cylA</i> positive (n = 72)	P ²	<i>gelE</i> negative (n = 17)	<i>gelE</i> positive (n = 117)	P ³	<i>ace</i> negative (n = 61)	<i>ace</i> positive (n = 73)	P ⁴
Ampicillin	11 (37.9)	26 (24.8)	0.16	17 (27.4)	20 (27.8)	0.96	8 (47.1)	29 (24.8)	0.06	18 (29.5)	19 (26.0)	0.65
Gentamicin	17 (58.6)	89 (84.8)	0.003	46 (74.2)	60 (83.3)	0.20	11 (64.7)	95 (81.2)	0.13	49 (80.3)	73 (100.0)	0.75
Chloramphenicol	6 (20.7)	49 (46.7)	0.01	23 (37.1)	32 (44.4)	0.39	3 (17.6)	52 (44.4)	0.05	27 (44.3)	28 (38.4)	0.49
Ciprofloxacin	18 (62.1)	92 (87.6)	0.002	47 (75.8)	63 (87.5)	0.08	12 (70.6)	98 (83.8)	0.19	52 (85.2)	58 (79.5)	0.39
Erythromycin	22 (75.9)	97 (92.4)	0.02	52 (83.9)	67 (93.1)	0.10	13 (76.5)	106 (90.6)	0.10	54 (88.5)	65 (89.0)	0.93
Rifampicin	13 (44.8)	54 (51.4)	0.53	28 (45.2)	39 (54.2)	0.30	8 (47.1)	59 (50.4)	0.80	31 (50.8)	36 (49.3)	0.86
Tetracycline	13 (44.8)	80 (76.2)	0.002	34 (54.8)	59 (81.9)	0.001	6 (35.3)	87 (74.4)	0.002	37 (60.7)	56 (76.7)	0.05

All P values determined using the χ^2 test. P values ≤ 0.05 are highlighted in bold.

¹ P values derived from *asa1* negative vs. *asa1* positive *E. faecalis* strains.

² P values derived from *cylA* negative vs. *cylA* positive *E. faecalis* strains.

³ P values derived from *gelE* negative vs. *gelE* positive *E. faecalis* strains.

⁴ P values derived from *ace* negative vs. *ace* positive *E. faecalis* strains.

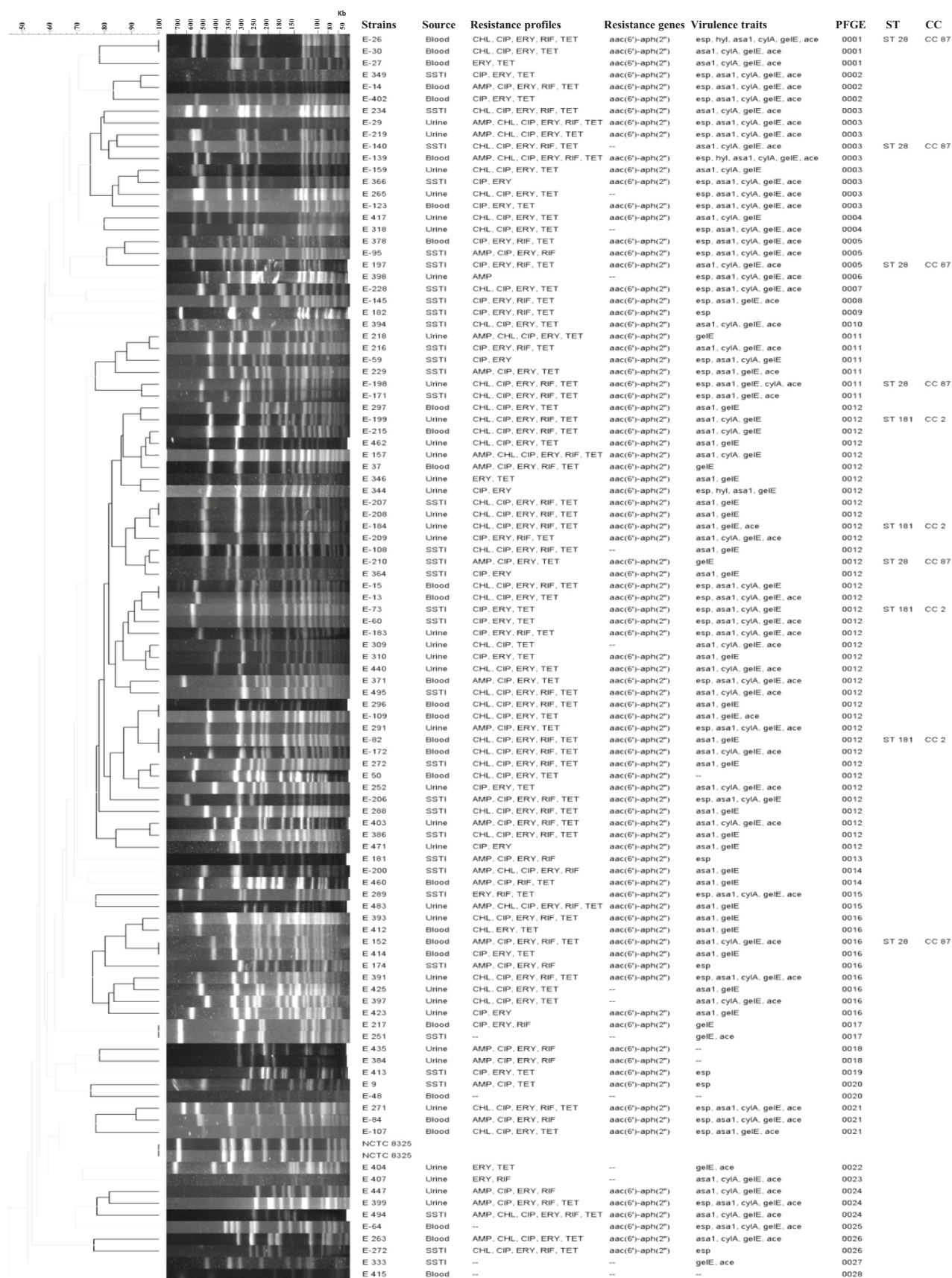


Fig. 1. Clonal relatedness of 103 *E. faecalis* isolates as based on *Sma*I macrorestriction analysis in PFGE and subsequent phylogenetic analysis (Dice coefficient using UPGMA clustering; InfoQuest™ FP Software v. 5-4 (Bio-Rad Laboratories, USA); settings: optimization 1.0%; tolerance 1.0%). PFGE patterns of *E. faecalis* isolates were delineated with a 75% similarity cut-off value (Tzavaras et al., 2012), and indicated by dark lines. Legend: AMP, ampicillin; CHL, chloramphenicol; CIP, ciprofloxacin; ERY, erythromycin; RIF, rifampicin; TET, tetracycline; *aac(6)-aph(2)*, PCR results for *aac(6)-aph(2)* genes; *esp*, *hyl*, *asa1*, *cya*, *gelE* and *ace*, PCR results for *esp* or *hyl* or *asa1* or *cya* or *gelE* or *ace* genes; SSTI, skin and soft tissue; PFGE, PFGE types; ST, MLST sequence type; CC, clonal complex.

Table 4
MLST, source, patient location, antibiotic resistance, virulence traits, diagnosis and outcome for selected strains of *E. faecalis* clones (*n* = 10).

Isolate	Year	ST	MLST alleles	CC	PFGE	Source	Patient location	Antibiotic resistance profiles	Antibiotic resistance gene	Virulence traits	Diagnosis	Outcome
E-73	2010	ST181	12-5-3-7-6-1-5	CC2	12	SSTI	Surgical ward	CIP, ERY, TET	<i>aac(6)-aph(2)</i>	<i>esp</i> , <i>asa1</i> , <i>cylA</i> , <i>gelE</i>	Aortic valve replacement	Discharged
E-82	2010	ST181	12-5-3-7-6-1-5	CC2	12	Blood	Surgical ward	CHL, CIP, ERY, RIF, TET	<i>aac(6)-aph(2)</i>	<i>asa1</i> , <i>gelE</i>	Posterior fossa mass with Hydrocephalus	Discharged
E-184	2011	ST181	12-5-3-7-6-1-5	CC2	12	Urine	Surgical ward	CHL, CIP, ERY, RIF, TET	<i>aac(6)-aph(2)</i>	<i>asa1</i> , <i>gelE</i> , <i>ace</i>	Carcinoma, UTI, Chronic Kidney disease	Discharged
E-199	2012	ST181	12-5-3-7-6-1-5	CC2	12	Urine	Surgical ward	CHL, CIP, ERY, RIF, TET	<i>aac(6)-aph(2)</i>	<i>asa1</i> , <i>cylA</i> , <i>gelE</i>	Renal transplantation, ADPKD	Discharged
E-26	2010	ST28	4-4-8-3-8-1-3	CC87	1	Blood	Medical ward	CHL, CIP, ERY, RIF, TET	<i>aac(6)-aph(2)</i>	<i>esp</i> , <i>hyl</i> , <i>asa1</i> , <i>cylA</i> , <i>gelE</i> , <i>ace</i>	Pemphigus vulgaris	Discharged
E-140	2011	ST28	4-4-8-3-8-1-3	CC87	3	SSTI	Surgical ward	CHL, CIP, ERY, RIF, TET	–	<i>asa1</i> , <i>cylA</i> , <i>gelE</i> , <i>ace</i>	Osteosarcoma, implant removal	Discharged
E-152	2011	ST28	4-4-8-3-8-1-3	CC87	16	Blood	ICU	AMP, CIP, ERY, RIF, TET	<i>aac(6)-aph(2)</i>	<i>esp</i> , <i>asa1</i> , <i>cylA</i> , <i>gelE</i> , <i>ace</i>	SAH, Left frontal temporal craniotomy	Discharged
E-197	2012	ST28	4-4-8-3-8-1-3	CC87	5	SSTI	Medical ward	CIP, ERY, RIF, TET	<i>aac(6)-aph(2)</i>	<i>asa1</i> , <i>cylA</i> , <i>gelE</i> , <i>ace</i>	Acute pancreatitis, Intra-abdominal sepsis	Discharged
E-198	2012	ST28	4-4-8-3-8-1-3	CC87	11	Urine	ICU	CHL, CIP, ERY, RIF, TET	<i>aac(6)-aph(2)</i>	<i>esp</i> , <i>asa1</i> , <i>cylA</i> , <i>gelE</i> , <i>ace</i>	Diabetic ketoacidosis	Discharged
E-210	2012	ST28	4-4-8-3-8-1-3	CC87	12	SSTI	ICU	AMP, CIP, ERY, TET	<i>aac(6)-aph(2)</i>	<i>gelE</i>	Infected endocarditis	Discharged

MLST, multi-locus sequence typing; ST, sequence type; CC, clonal complex; PFGE, pulsed-field gel electrophoresis; SSTI, skin and soft tissue; ICU, intensive care unit; AMP, ampicillin; CHL, chloramphenicol; CIP, ciprofloxacin; ERY, erythromycin; RIF, rifampicin; TET, tetracycline; *aac(6)-aph(2)*, PCR results for *aac(6)-aph(2)* genes; *esp*, *hyl*, *asa1*, *cylA*, *gelE* and *ace*, PCR results for *esp* or *hyl* or *asa1* *cylA* or *gelE* or *ace* genes; SAH, subarachnoid hemorrhage; UTI, urinary tract infection; ADPKD, autosomal dominant polycystic kidney disease.

et al., 2019). In addition, there was a significant association between virulence genes-*asa1*, *cylA*, and *gelE* and antibiotics such as tetracycline, chloramphenicol and high-level gentamicin due to the presence of mobile resistance and virulence determinants on conjugative transposon, either on chromosomal DNA or plasmid, resulting in co-transfer between *E. faecalis* isolates during the conjugation. Thus, the presence of multiple virulence determinants in clinical isolates of *E. faecalis* and its association with antibiotic resistance make this organism fit to cause serious nosocomial infections with increased morbidity and mortality.

In this study, MLST and PFGE were chosen to examine and determine whether particular *E. faecalis* lineages or clones have been associated with infections in our hospital. As far as we know, this is the first study to focus on the molecular epidemiology of clinical isolates of MDR *E. faecalis* from India. Our results show the dissemination of majority of genetically related HLGR *E. faecalis* isolates with four major PFGE clones. Interestingly, these four major PFGE clones have been associated with MDR activity and carried virulence determinants in most of their isolates, suggesting that all these factors may be responsible for their selection and persistence in our hospital. However, we reported some minor PFGE clones possibly because of the presence and exchange of mobile genetic elements containing resistance and virulence genes between MDR *E. faecalis* strains. Similar to our PFGE findings, clinical *E. faecalis* isolates have been identified as very polyclonal with wide diverse populations in previous reports from Italy (Zarrilli et al., 2005) and Australia (Coombs et al., 2014) due to the extensive plasticity of *E. faecalis* genome, which often acquires genetic elements by conjugation.

Based on MLST, Ruiz-Garbajosa et al. reported the seven most prevalent *E. faecalis* clones namely ST6, ST9, ST16, ST21, ST28, ST40, and ST87 among clinical and outbreak-associated isolates (Ruiz-Garbajosa et al., 2006). Previous studies have shown that *E. faecalis* STs belong to six CCs (CC2, CC16, CC21, CC30, CC40, and CC87) have played a predominant role in the spread of antimicrobial resistance in hospitals and have contributed to higher resistance rates. Three CCs (CC2, CC16, and CC87) were significantly associated with the MDR phenotype (McBride et al., 2007). Another study showed that CC2, CC9, and CC87 belong to high-risk enterococcal CCs (HiRECCs) i.e. CCs enriched with MDR comprising colonizing and invasive isolates from hospitalized patients (Ruiz-Garbajosa et al., 2006; Kuch et al., 2012).

The emergence of ST28 (SLV of 87) and ST181 (SLV of ST6) belonging to HiRECCs CC87 and CC2 indicates and alerts us to the presence and spread of global MDR *E. faecalis* clones in our hospital. In previous studies, authors have been reported these STs or its single locus variants (SLV) such as ST87 and ST6 from Poland, Europe, USA, Malaysia, and Iran (Kawalec et al., 2007; Kuch et al., 2012; McBride et al., 2007; Weng et al., 2013; Zalipour et al., 2019). Although none of the vancomycin-resistant *E. faecalis* were detected in our study, the presence of two major CCs (CC2 and CC87) associated with these MDR strains circulating in our hospital wards was most worrying because vancomycin-susceptible isolates belonging to these CCs showed the ability to rapidly acquire vancomycin resistance gene, as previously reported in enterococci and staphylococci (Kawalec et al., 2007; Rice et al., 2018). Similarly, in our previous 5-year study, we reported decreased susceptibility to vancomycin by MIC testing of methicillin-resistant *Staphylococcus aureus* (Dhawan et al., 2010). The interspecies transmission of vancomycin resistance may be possible. Increased virulence determinants and resistance to antimicrobial agents in *E. faecalis* significantly linked to the selection and expansion of certain CCs like CC2 and CC87 spreading in hospitalized patients. In addition, the diversity of the antibiogram and virulence gene profiles of CC2 and CC87 isolates from our hospital settings further supports the suggestion that CC2 and CC87 are competent in the exchange of DNA. The continued use of antimicrobial agents in hospitals, in the community, and in animal husbandry will further promote the selection and dissemination of MDR clones. Therefore, ongoing surveillance and improved procedures for the control of infections in hospitals are

indispensable to curb further dissemination of MDR *E. faecalis* strains.

Our study was limited to the number of isolates analyzed by MLST and observations derived from a single hospital. However, we believe that this is the first report from India which provided a valuable insight into the molecular epidemiology of *E. faecalis* strains circulating in an Indian hospital. In summary, results obtained in this study show that the high prevalence of MDR and virulence determinants in clinical isolates limits therapeutic options against infection caused by *E. faecalis* strains. Particularly, the high prevalence of HLGR and ampicillin eliminates the use of combination of aminoglycoside and beta-lactam antibiotics for empirical treatment against infection caused by this bacterium. The MDR-associated hospital *E. faecalis* is significantly attached to CC2 and CC87. Clonal expansion of the hospital-associated worldwide-disseminated CC2 and CC87 are significantly associated with antibiotic resistance and virulence determinants in our hospital. These findings suggest close monitoring of the epidemiology of these strains and the need to maintain a judicious antibiotic use policy.

Disclosure statement

All authors report that they have no conflicts of interest relevant to this study. The manuscript has been read and approved by all authors.

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Author statement

Chandrabhan Rao: Conceptualization, Methodology, Laboratory work, Sample collection, Validation, Resources, Project administration, Implementation of softwares for PFGE and MLST data analysis, Data curation, Data analysis, Writing- Original draft preparation, Visualization, Investigation, Writing- Reviewing and Editing.

Benu Dhawan: Conceptualization, Supervision, Funding acquisition, Resources, Project administration, Provided Lab facilities, helped in Sample collection, Visualization, Data curation, helped in Writing- Original draft preparation, Writing- Reviewing and Editing.

Sreenivas Vishnubhatla: Co-supervision, Data analysis, helped in Data curation, Data Verification, Supported in visualization, Reviewing and Editing.

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